INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

# FRAME, FIELD PACK, (RIVETED), AND SHELF, CARGO SUPPORT (LIGHTWEIGHT), LC-1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers a lightweight aluminum pack frame and a detachable aluminum cargo support shelf (see 6.1).

1.2 Classification. The pack frame and cargo support shelf shall be classified as follows (see 6.2):

Item 1 - Pack frame (riveted)
Item 2 - Cargo support shelf

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8465

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SPECIFICATIONS

FEDERAL

| | |
|---|---|
| A-A-50083 | - Bag, Plastic, Folded Garment |
| QQ-A-200/8 | - Aluminum Alloy 6061, Bar, Rod, Shapes, Tube and Wire, Extruded |
| QQ-A-200/9 | - Aluminum Alloy 6063, Bar, Rod, Shapes, Tube and Wire, Extruded |
| QQ-A-250/8 | - Aluminum Alloy 5052, Plate and Sheet |
| QQ-A-250/11 | - Aluminum Alloy 6061, Plate and Sheet |
| QQ-S-763 | - Steel Bars, Wire, Shapes, and Forgings, Corrosion Resisting |
| TT-E-527 | - Enamel, Alkyd, Lusterless, Low Voc Content |
| TT-P-1757 | - Primer Coating, Zinc Chromate, Low-Moisture Sensitivity |
| WW-T-700/6 | - Tube, Aluminum Alloy, Drawn, Seamless, 6061 |
| PPP-B-636 | - Boxes, Shipping, Fiberboard |

MILITARY

| | |
|---|---|
| MIL-C-5541 | - Chemical Conversion Coatings on Aluminum and Aluminum Alloys |
| MIL-A-8625 | - Anodic Coatings, for Aluminum and Aluminum Alloys |
| MIL-C-13924 | - Coating, Oxide, Black, for Ferrous Metals |
| MIL-L-35078 | - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For |

STANDARDS

FEDERAL

| | |
|---|---|
| FED-STD-595 | - Colors Used in Government Procurement |

MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-129 | - Marking for Shipment and Storage |
| MIL-STD-147 | - Palletized Unit Loads |
| MIL-STD-731 | - Quality of Wood Members for Containers and Pallets |

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)



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* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

| | |
|---|---|
| 2-3-290 | - Frame, Pack, Ground Troops, Lightweight, LC-1; Assembly |
| 2-3-291 | - Shelf, Cargo Support, for Frame, Pack, Ground Troops, Lightweight, LC-1 |
| 2-3-493 | - Strap Packing Configuration |

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-UX, Natick, MA 01760-5017.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

| | |
|---|---|
| B 244 | - Measurement of Thickness of Anodic Coatings on Aluminum and of Other Nonconductive Coatings on Nonmagnetic Basis Metals with Eddy-Current Instruments |
| B 633 | - Electrodeposited Coatings of Zinc on Iron and Steel |
| D 1005 | - Measurement of Dry-Film Thickness of Organic Coatings Using Micrometers |
| D 3951 | - Standard Practice for Commercial Packaging |

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

* 3.2 Materials and components. Materials and components shall be as specified herein and on applicable drawings. Materials and components not definitely specified shall be of the quality normally used by the manufacturer provided the completed item complies with all provisions of this specification. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.2.1 Aluminum alloys.

3.2.1.1 Tube. Aluminum alloy tube shall conform to 6061-T6 of WW-T-700/6.

3.2.1.2 Sheet. Aluminum alloy sheet shall conform to 5052-H32 or 5052-H34 of QQ-A-250/8 or 6061-T6 of QQ-A-250/11, as applicable.

3.2.1.3 Shapes. Aluminum alloy shapes (channel) shall conform to 6061-T6 of QQ-A-200/8 or 6063-T5 of QQ-A-200/9.

3.2.2 Wire, corrosion-resisting steel. Corrosion-resisting steel wire shall conform to class 302 or 304, condition B, cold drawn finish of QQ-S-763.

3.2.3 Rivets. Rivets shall be as shown on Drawing 2-3-290. The zinc plating of the steel semi-tubular rivets (Nos. 14, 27, 28, 30, 31, 34 and 35) shall conform to type II, SC-3 of ASTM B 633.

3.2.4 Primer. Primer shall conform to composition G or L, as applicable, color Y of TT-P-1757.

3.2.5 Enamel. Enamel shall conform to composition G or L of TT-E-527, as applicable, and green color No. 34087 of FED-STD-595.

3.2.6 Alternate components. Components offered as equivalent to components specified herein and on the referenced drawings as specific manufacturer's equal shall be functionally equal with the manufacturer's part number identified. The incorporation of such a component in the design of the specified end product shall not require modification to any components unless such modification is specifically approved by the contracting officer. Prior to commencing production, the contractor shall submit for the contracting officer's approval, a list identifying each alternate component, together with proof that each listed component complies with requirements specified herein. The contracting officer, at his option, may require a physical sample of the proposed substitution. Approval of the submitted listing, together with necessary supporting data, does not relieve the contractor of the responsibility that these components perform in accordance with specified requirements when incorporated into the end product.



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3.3 Construction.

3.3.1 Item 1 - pack frame. The construction of the pack frame shall conform to Drawing 2-3-290 and as specified herein.

3.3.1.1 Body and insert. The body and insert shall be fabricated of aluminum alloy 6061-T6 drawn tube as specified in 3.2.1.1. At the option of the contractor, extruded tube conforming to 6061-T6 of QQ-A-200/8 may be used as an alternate providing the difference between the inside diameter of the frame body and the outside diameter of the insert does not exceed 0.024 inch when tested as specified in 4.4.2.2.

3.3.1.2 Cross members and vertical braces. The cross members and vertical braces shall be fabricated of aluminum alloy 5052-H32 sheet as specified in 3.2.1.2.

3.3.1.3 Loop retainer and web guide. Loop retainer and web guide shall be fabricated of aluminum alloy 5052-H34 sheet as specified in 3.2.1.2.

3.3.1.4 Strap, center. The center strap shall be fabricated of either aluminum alloy 5052-H32 or 6061-T6, as specified in 3.2.1.2.

3.3.1.5 Brace. The channel brace shall be fabricated by either of the following:

a. Aluminum alloy 6061-T6 or 6063-T5 extruded channel as specified in 3.2.1.3.

b. Aluminum alloy 5052-H32 sheet as specified in 3.2.1.2 and formed to the dimensions shown on Drawing 2-3-290.

3.3.1.6 Loop. The loop shall be fabricated of corrosion-resisting steel wire specified in 3.2.2.

3.3.2 Item 2 - cargo support shelf. The construction of the cargo support shelf shall conform to Drawing 2-3-291 and as specified herein. The cargo support shelf shall be fabricated of aluminum alloy 5052-H34 sheet as specified in 3.2.1.2. The cargo support shelf shall be capable of being fitted on the frame without any looseness (see 6.4).

* 3.3.3 Riveting. Final sizing of rivet holes shall be accomplished at time of assembly to preclude misalignment of rivet holes. Rivet holes shall be drilled in the tubular body of frame. In lieu of drilling, punching or piercing of holes shall be permitted providing the diameter of the tubular body is free of any deformation exceeding the limits of plus or minus 0.010 inch and the inner wall of tubular body is free of any burr exceeding 0.010 inch when examined as specified in 4.4.2.1. Mating surfaces or riveting components shall be drawn up tightly and the rivet head shall be in contact with its mating surface. Rivets shall be properly set and free of any looseness. Roll of a rivet shall be free



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from more than one split. Set rivets shall be free of any portion of a rivet set or rolled on itself in lieu of being set or rolled on mating surface of component usually due to an enlarged or elongated hole. There shall be no dimpling of the outer tube for assembly of frame prior to or during the riveting process.

3.4. Finish.

3.4.1 Pack frame. After forming, but prior to assembly, aluminum components (frame body, cross members, strap, loop retainer, web guide, channel and vertical braces) shall be hardcoat anodized to conform to type III, 0.001 inch minimum thickness, class 2, of MIL-A-8625. The color shall be green, No. 34079 or 34086 of FED-STD-595. The final finish shall be lusterless produced by means of appropriate chemical etching preparatory to hardcoat anodizing. The anodic coating shall be uniform in appearance, continuous, smooth, adherent, and free from powdery area and loose film break. The contractor shall repair any anodic coating damaged during examination in accordance with class 1A requirements of MIL-C-5541. After forming, the corrosion-resisting steel wire loop shall be given a black oxide coating conforming to class 4 of MIL-C-13924.

3.4.1.1 Pack frame, epoxy powder coated. The epoxy coated pack frame may be used in lieu of hardcoat anodized pack frame finish specified in 3.4.1. The assembled frame (without the plastic grommets) shall be prepared and electrodeposit epoxy coated as specified in 3.4.2.4. Upon completion of the epoxy coated pack frame finish, the plastic grommets shall be assembled to the finished frame.

3.4.2 Cargo support shelf.

3.4.2.1 Surface preparation. All surfaces of the cargo support shelf shall be prepared for painting in accordance with class 1A treatment of MIL-C-5541. Treated surfaces shall be kept clean during subsequent handling, being handled only by clean-gloved hands or by clean implements.

3.4.2.2 Priming. Prepared surfaces shall be coated with the primer specified in 3.2.4.

3.4.2.3 Enameling. Primed surfaces shall be coated with the enamel specified in 3.2.5. The final finish shall be smooth and uniform in appearance, free of any sags, runs, pinholes, blisters, wrinkles, frothing, or streaks.

3.4.2.4 Epoxy coating. As an option, in lieu of the priming as specified in 3.4.2.2 and enameling as specified in 3.4.2.3, the prepared surfaces may be prepared and coated as follows:

a. The cargo shelf shall be prepared by mechanical means. The surfaces shall be thoroughly and uniformly textured by sand blasting with 80 grit aluminum oxide or 80 grit steel. Prepared surfaces shall be kept clean during the subsequent handling. Handling to be



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by clean-gloved hands or by clean implements. The surfaces shall be retained in a clean environment. The surface shall be epoxy coated not more than 8 hours after sand blasting.

b. The finish coating shall be lusterless green (olive green) No. 34079, 34086, or 34087 of FED-STD-595, electrodeposited epoxy powder coating conforming to H.B. Fuller Co., Kativo Powder, IF-3561 or equal.

c. The electrodeposited epoxy powder coating shall be baked in accordance with commercial practice.

d. The thickness of the baked coating shall be a minimum of 0.0025 inch when tested as specified in 4.4.3.4.

e. The final coating shall be smooth and uniform in appearance, free from wrinkles and streaks.

3.5 Marking for identification. Prior to finishing as specified in 3.4, the frame and shelf shall be indent stamped in the location as shown in applicable drawing. After finishing, the marking shall be legible.

3.6 Workmanship. The frame and shelf shall be free from burrs, rough or sharp edges (including rivet), malformations, deformations, cracks, punctures, fractures, nonspecified holes, enlarged or elongated holes, and misalignment of holes and shall not be twisted or bent out of shape. The finish shall be free from any area of no coating and exposed bare metal surface due to scratch, scuff, dig, or abrasion. The finished pack frame and cargo support shelf shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.



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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3.1, 4.4.3.2, 4.4.3.3, and when the optional epoxy powder coating method is used, 4.4.3.4. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 In-process inspection.

4.4.2.1 Operation examination. Examination shall be made of the following operations to establish conformance to specified requirements. Whenever nonconformance is noted, correction shall be made to the applicable process and all items processed.

a. Examination of tubular body of frame. When rivet holes are punched or pierced in lieu of drilling, the tubular body of frame shall be free from any diameter deformation exceeding the limits of plus or minus 0.010 inch and the inner wall of tubular body of



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frame shall be free of any burr exceeding 0.010 inch as specified in 3.3.3.

b. Finish operation examination. The aluminum frame components shall be examined to determine that the appropriate chemical etching is performed prior to the application of the hardcoat anodizing (see 3.4.1). The aluminum shelves shall be examined at appropriate stages to determine compliance with the surface preparation requirement of 3.4.2.1 and priming requirement of 3.4.2.2 prior to enameling, as applicable.

4.4.2.2 Difference in diameters examination. When extruded tube is used in lieu of drawn tube, the difference between the inside diameter of the frame body and the outside diameter of the insert shall be dimensionally examined as specified in 3.3.1.1. The lot shall be the number of formed (bent) frame bodies prior to assembly with the accompanying inserts offered for inspection at one time. The sample unit shall be one frame body with two inserts. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0. The inside diameter of each end of the frame body and the outside diameter of each insert shall be measured to the nearest 0.001 inch. Any difference between inside diameter of frame body and outside diameter of insert dimensions exceeding 0.024 inch shall be classified as a defect.

4.4.3 End item inspection.

4.4.3.1 Pack frame visual examination. The end items shall be examined for the defects listed in table I. The lot size shall be expressed in units of pack frames. The sample unit shall be one pack frame. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 4.0 for major defects and 10 for total (major and minor combined) defects.

TABLE I. End item (pack frame) visual defects

| Examine | Defect | Classification Major | Minor |
|---|---|---|---|
| Finish | Any component not finished | 101 | |
| | Color of finish is not as specified | | 201 |
| | Anodic coating not uniform in appearance, not continuous, not smooth, not adherent, or evidence of powdery area or loose film break | | 202 |
| | Area of no coating | | 203 |



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TABLE I. End item (pack frame) visual defects (cont'd)

| Examine | Defect | Classification Major | Classification Minor |
|---|---|---|---|
| Finish (cont'd) | | | |
| | Exposed bare metal surface due to scratch, scuff, dig, or abrasion | | 204 |
| | NOTE: Defects in anodic, enamel and epoxy finish attributable to handling during examination will not be scored as a defect. However, the contractor shall touch up areas of finish damaged during examination. Any anodic coating damaged during examination shall be repaired in accordance with class 1A requirements of MIL-C-5541. | | |
| Construction and workmanship | Any component missing | 102 | |
| | Not assembled as specified | 103 | |
| | Burrs or rough edge (including rivet) | | 205 |
| | Sharp edges (including rivet), malformations, deformations, cracks, punctures, or fractures | 104 | |
| | Nonspecified holes, enlarged or elongated holes, misalignment of holes; component twisted or bent out of shape | | 206 |
| Riveting | Mating surfaces of riveted components not drawn up tightly | | 207 |
| | Rivet head not in contact with its mating surface | | 208 |
| | Rivet not properly set (i.e., is loose) | | 209 |
| | Roll of a rivet contains more than one split | | 210 |
| | Portion of a rivet is set or rolled on itself in lieu of being set or rolled on mating surface of component | | 211 |
| | Dimpling of the outer tube | 105 | |
| Identification marking | Missing, illegible, misplaced, incorrect, or not indent stamped | | 212 |



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4.4.3.2 Cargo support shelf visual examination. The end items shall be examined for the defects listed in table II. The lot size shall be expressed in units of cargo support shelves. The sample unit shall be one cargo support shelf. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 4.0 for major defects and 10 for total (major and minor combined) defects.

TABLE II. End item (cargo support shelf) visual defects

| Examine | Defect | Classification Major | Minor |
|---|---|---|---|
| Finish | Missing | 101 | |
| | Color of finish is not as specified | | 201 |
| | Coating is not smooth or not uniform in appearance | | 202 |
| | Evidence of any sags, runs, pinholes, blisters, wrinkles, frothing, or streaks | | 203 |
| | Area of no coating | | 204 |
| | Exposed bare metal due to scratch, scuff, dig, or abrasion | | 205 |
| Construction and workmanship | Shelf is not capable of being fitted on the frame or evidence of looseness (see 6.4) | 102 | |
| | Burrs or rough edge | | 206 |
| | Sharp edges, malformations, deformations, cracks, punctures, or fractures | 103 | |
| | Nonspecified hole; component twisted or bent out of shape | | 207 |
| Identification marking | Missing, illegible, misplaced, incorrect, or not indent stamped | | 208 |

* 4.4.3.3 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified on the drawings that can be determined on the end items without damaging or disassembling the end items. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of frames or shelves. The sample unit shall be one frame or shelf, as applicable. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 10.



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4.4.3.4 Epoxy coating thickness examination (when the optional epoxy powder coating is used). The end items shall be examined for thickness of the epoxy coating in accordance with ASTM D 1005 or ASTM B 244. Three observations shall be made per sample unit. Any epoxy coating thickness less than specified in 3.4.2.4 shall be classified as a defect. The lot size shall be expressed in units of frames or shelves. The sample level shall be one frame or shelf, as applicable. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.4 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

| Examine | Defect |
|---|---|
| Marking (exterior and interior) | Omitted; incorrect; illegible; of improper size, location, sequence, or method of application |
| Materials | Any component missing, damaged, or not as specified |
| Workmanship | Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling<br>Bulged or distorted container |
| Content | Number per container is more or less than required |

4.4.5 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

| Examine | Defect |
|---|---|
| Finished dimensions | Length, width, or height exceeds specified maximum requirement |
| Palletization | Pallet pattern not as specified<br>Load not bonded as specified |



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| | |
|---|---|
| Weight | Exceeds maximum load limits |
| Marking | Omitted; incorrect; illegible; of improper size, location, sequence, or method of application |

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A preservation.

* 5.1.1.1 Unit packaging. Each cargo support shelf strap shall be unit packed in a polyethylene bag in accordance with A-A-50083.

* 5.1.2 Commercial preservation. Cargo support shelves shall be preserved in accordance with ASTM D 3951.

* 5.2 Packing. Packing shall be level A or Commercial, as specified (see 6.2).

5.2.1 Level A packing.

* 5.2.1.1 Pack frame. Four pack frame assemblies, reverse nested and separated by a fiberboard divider pad as shown in drawing 2-3-493, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each box shall be fitted with a box liner conforming to type CF, class weather resistant, variety DW, grade V15c of PPP-B-636. Inside dimensions of the box shall be approximately 23-1/2 inches in length, 13-1/2 inches in width, and 11 inches in depth. Approximate dimensions are furnished as a guide only. The divider pad shall be the full length and width of the box and constructed from corrugated fiberboard. Each container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced in accordance with the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

* 5.2.1.2 Cargo support shelves. Fifty cargo support shelves, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Level A intermediate unit packs shall be packed flat, one in length and five in depth. Inside dimensions of each box shall approximate 12-1/2 inches in length, 12-1/2 inches in width, and 13 inches in depth. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4. Shipping containers



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shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing.

* 5.2.2.1 Pack frame. Four pack frame assemblies, reverse nested and separated by a fiberboard divider pad as shown in drawing 2-3-493, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each box shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Inside dimensions of the box shall be approximately 23-1/2 inches in length, 13-1/2 inches in width, and 11 inches in depth. Approximate dimensions are furnished as a guide only. The pad shall be the full length and width of the box and constructed with corrugated fiberboard. Each box shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4.

* 5.2.2.2 Cargo support shelf. Fifty cargo support shelves, preserved as specified in 5.1, shall be packed in a snug-fitting, fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 200 of PPP-B-636. Level A unit packs shall be packed flat, one in length and five in depth. Inside dimensions of the box shall be approximately 12-1/2 inches in length, 12-1/2 inches in width, and 13 inches in depth. Each shipping container shall be closed in accordance with method II of the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4.

5.2.2.3 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4.

* 5.2.3 Commercial packing. Pack frames or cargo support shelves, preserved as specified in 5.1, or strap and pack frame assembly shall be packed in accordance with ASTM D 3951.

* 5.3 Palletization. When specified (see 6.2), pack frames or cargo support shelves packed as specified in 5.2.2 or 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be number 91 for pack frames and number 97 for cargo support shelves in accordance with the appendix of MIL-STD-147.

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* 5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The pack frame is intended for use as a mount for a pack, which is in turn used by personnel to carry items of clothing and equipment. The frame may be used either with or without the cargo support shelf.

* 6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Item required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3, and 6.3).
e. Levels of preservation and packing (see 5.1 and 5.2).
f. Type and class of unit load required (see 5.2.1.1 and 5.2.1.2).
g. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.3).
h. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Support shelves. When support shelves are procured without pack frames, the contracting officer shall furnish the contractor with an item 1 (pack frame) to determine proper fit (see table II).

6.5 International standardization agreement. Certain provisions of this document are the subject of international standardization agreement as cited in NATO, STANAG No. 2311. When amendment, revision, or cancellation of this document is proposed which will effect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.



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* 6.6 Subject term (key word) listing.

Equipment, outdoor
Form
Framework
Holder
Skeletal structure

6.7 Changes from previous issue. The margins of this document are marked with asterisks (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - NU
Air Force - 99

Preparing activity:

Army - GL

(Project 8465-0035)

Review activities:

Army - MD
Navy - MC
Air Force - 82
DLA - CT

User activities:

Navy - YD
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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|
| | MIL-F-43834E | 1990 December 14 |

3. DOCUMENT TITLE
Frame, Field Pack, (Riveted), And Shelf, Cargo Support (Lightweight), LC-1

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
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| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON<br>(If applicable) | 7. DATE SUBMITTED (YYMMDD) |

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